

Title

Changes in circulating neurotrophic factors following psychological intervention in patients with generalized anxiety disorder

Abstract

Background: Whether psychotherapy-induced clinical improvement is accompanied by changes in neurobiological biomarkers is unclear. **Methods:** 82 adults with generalized anxiety disorder (mean age 41 years) were randomized to acceptance-based cognitive-behavioral group therapy (n=41) or supportive counseling (n=41) delivered over 14 weeks. Plasma neurotrophic factor levels and anxiety severity (Hamilton Anxiety Scale) were measured at baseline and post-treatment. **Results:** Both treatment groups showed significant increases in plasma neurotrophic factor from baseline to post-treatment (supportive therapy: mean increase 2031.4 pg/mL, $t=4.79$, $p<0.001$; acceptance-based therapy: mean increase 1963.2 pg/mL, $t=4.42$, $p<0.001$). Anxiety symptoms decreased in both groups. However, changes in plasma neurotrophic factor did not correlate with reductions in anxiety symptom severity in either treatment modality. **Conclusions:** Both active psychological treatments were associated with increases in circulating neurotrophic factor levels; however, the absence of correlation with symptom improvement suggests neurotrophic factor elevation may represent a general response to treatment engagement rather than a specific mechanistic pathway. These findings should be interpreted cautiously and do not establish a direct link between biomarker changes and clinical outcomes.